PRIME-LEVEL CONVERSE THEOREM WITH TWISTS OF MODULUS BELOW THE SQUARE ROOT

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The following argument recovers modularity using character-sum twists of modulus below the square root of a prime level. No Euler-product hypothesis is needed for the linear converse theorem; the Euler product and its ramified factor give the newness assertion in Corollary 5.

Write $e(x) = \exp(2\pi i x)$ and

$$T_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \qquad W_M = \begin{pmatrix} 0 & -1 \\ M & 0 \end{pmatrix}, \qquad w_p = p^{-1/2} W_p.$$

We use the determinant-normalized slash operator

$$(F|_k A)(z) = (\det A)^{k/2}(cz+d)^{-k} F(Az) \quad \left(A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},\ \det A > 0 \right).$$

Positive scalar matrices act trivially. Throughout, p is a prime, k is a positive even integer, and $Q = \lfloor \sqrt{p} \rfloor$. Since a prime is not a square, $q < \sqrt{p}$ and $q \le Q$ are equivalent for integer q.

Theorem 1. *Let $(f_n)_{n\ge 1}$ be a sequence of polynomial growth and put*

$$F(z) = \sum_{n\ge 1} f_n n^{(k-1)/2} e(nz).$$

For every $1 \le q < \sqrt{p}$ and every Dirichlet character χ modulo q, including the imprimitive characters, put

$$c_\chi(n) = \sum_{a \in (\mathbb{Z}/q\mathbb{Z})^\times} \chi(a) e(an/q), \qquad \Lambda_f(s, c_\chi) = (pq^2)^{(s-1/2)/2} \Gamma_{\mathbb{C}}\left(s + \frac{k-1}{2} \right) \sum_{n\ge 1} \frac{f_n c_\chi(n)}{n^s},$$

where $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$. Suppose these functions extend to entire functions of finite order and, for one fixed $\epsilon \in \{1, -1\}$, satisfy

$$\Lambda_f(s, c_\chi) = \epsilon\, \overline{\chi(-p)}\, \Lambda_f(1-s, c_{\bar\chi}). \tag{1}$$

Then $F \in S_k(\Gamma_0(p))$ and $F|_k W_p = \epsilon i^{-k} F$.

For modulus 1, the sole character and its sum are both 1; thus the untwisted functional equation is included. If one defines c_χ using $\bar\chi(a)$ instead of $\chi(a)$, the factor in (1) is $\chi(-p)$ instead. These are equivalent conventions, but should not be mixed.

1. An elementary height-generation lemma

The following formulation isolates the geometric input. The lower-row finiteness condition is deliberately weaker than a description of a fundamental polygon.

Lemma 2 (Height generation)**.** *Let $H \le G \le \mathrm{SL}_2(\mathbb{R})$ and suppose:*

1. *$T_1 \in H$, and every element of G fixing ∞ belongs to H;*
2. *for each $z \in \mathbb{H}$ and each $B > 0$, there are only finitely many lower rows (c, d) of elements of G satisfying $|cz + d| \le B$;*

(3) *there is $c_0 > 0$ such that every nonzero lower-left entry of an element of G has absolute value at least c_0;*
(4) *there are finitely many $h_j \in H$, with lower rows (c_j, d_j) and $c_j \neq 0$, whose open isometric intervals*

$$\{x \in \mathbb{R} : |c_j x + d_j| < 1\}$$

cover $[0,1]$.

Then $H = G$.

Proof. For each fixed $z = x + iy \in \mathbb{H}$, the identity

$$\operatorname{Im}(gz) = \frac{y}{|cz + d|^2}$$

and condition (2) show that the set of heights in Hz greater than or equal to y is finite. It is nonempty because the identity belongs to H. Hence the height attains its maximum on the whole orbit. Choose a maximum-height point, and then apply a power of T_1 so that its real part lies in $[0,1]$. Denote the resulting point by $z' = x' + iy'$. Maximality implies $|c_j z' + d_j| \geq 1$ for each j; equivalently

$$y'^2 \geq c_j^{-2} - (x' + d_j/c_j)^2.$$

The continuous function

$$m(x) = \max_j \bigl(c_j^{-2} - (x + d_j/c_j)^2\bigr)$$

is positive on $[0,1]$ by condition (4). Compactness gives $\eta^2 = \min_{x \in [0,1]} m(x) > 0$. Thus every H-orbit has a representative of height at least η.

Now let $g \in G$ be arbitrary and take $Y > 1/(c_0^2 \eta)$. Apply the preceding conclusion to $g(iY)$ and choose $h \in H$ with $\operatorname{Im}(hg(iY)) \geq \eta$. If the lower-left entry c of hg were nonzero, then

$$\operatorname{Im}(hg(iY)) = \frac{Y}{c^2 Y^2 + d^2} \leq \frac{1}{c_0^2 Y} < \eta,$$

a contradiction. Therefore hg fixes ∞, so $hg \in H$ by condition (1), and $g \in H$. □

Lemma 3 (Small-denominator generators)**.** *Let*

$$G_p = \langle \Gamma_0(p), w_p \rangle.$$

Then G_p is generated by T_1, w_p, and all the matrices

$$\gamma_{q,a,b} = \begin{pmatrix} (1 + pab)/q & a \\ pb & q \end{pmatrix}, \qquad 1 \leq q \leq Q, \quad a, b \in \mathbb{Z}, \quad pab \equiv -1 \pmod{q}. \tag{2}$$

Proof. Let H be the subgroup generated by the displayed elements. Conjugation by w_p normalizes $\Gamma_0(p)$ and $w_p^2 = -I$; therefore

$$G_p = \Gamma_0(p) \ \cup \ \Gamma_0(p) w_p.$$

For $\gamma = \left(\begin{smallmatrix} a & b \\ pc & d \end{smallmatrix}\right) \in \Gamma_0(p)$, the lower row of γ is (pc, d), whereas that of γw_p is $(d\sqrt{p}, -c\sqrt{p})$. These are two discrete lattices of lower rows. For fixed $z = x + iy$ and $B > 0$, the inequality $|uz + v| \leq B$ bounds $|u| \leq B/y$ and $|v| \leq B + |ux|$, so only finitely many such rows occur. Every nonzero lower-left entry has absolute value at least $\sqrt{p}$.

An element of $\Gamma_0(p)$ fixing ∞ is $\pm T_1^n$. An element γw_p could fix ∞ only if $d = 0$, which would give $-pbc = 1$ and is impossible. Thus the stabilizer of ∞ in G_p is $\{\pm T_1^n : n \in \mathbb{Z}\} \subset H$.

It remains to check the interval cover. For every reduced b/q with $1 \leq q \leq Q$, the congruence $pab \equiv -1 \pmod{q}$ has a solution a, because $q < p$ and p is prime. The element $\gamma_{q,a,b} w_p \in H$ has lower row

$$(q\sqrt{p}, -b\sqrt{p}),$$

so its open isometric interval is

$$\left(\frac{b}{q}-\frac{1}{q\sqrt{p}},\ \frac{b}{q}+\frac{1}{q\sqrt{p}}\right). \tag{3}$$

Take just the fractions in the Farey sequence of order Q on $[0,1]$, including $0/1$ and $1/1$. Consecutive fractions $b/q < b'/q'$ satisfy

$$\frac{b'}{q'}-\frac{b}{q}=\frac{1}{qq'}, \qquad q+q'>Q.$$

Consequently $q+q' \geq Q+1 > \sqrt{p}$, and the sum of the two radii is

$$\frac{1}{q\sqrt{p}}+\frac{1}{q'\sqrt{p}}=\frac{q+q'}{qq'\sqrt{p}}>\frac{1}{qq'}.$$

The adjacent intervals overlap strictly and cover $[0,1]$. Applying Lemma 2 with $c_0=\sqrt{p}$ gives $H=G_p$. □

Remark 4. The proof also covers $p=2,3$, when $Q=1$ and the two Farey endpoints suffice. Strict overlap matters: the argument uses $Q+1>\sqrt{p}$, not $Q>\sqrt{p}$. Primality is used to make every denominator $q \leq Q$ invertible modulo p. No estimate of the index or unstated claim about generators is needed.

2. Recovering the individual transformation laws

Put

$$F_\chi(z)=\sum_{a\in(\mathbb{Z}/q\mathbb{Z})^\times}\chi(a)F(z+a/q), \qquad \eta=\epsilon i^{-k}\in\{1,-1\}.$$

The paired form of Hecke's Mellin-inversion argument applied to (1) gives

$$F_\chi|_k W_{pq^2}=\eta\,\overline{\chi(-p)}\,F_{\bar\chi}. \tag{4}$$

For completeness, if $M=pq^2$ and $\lambda_\chi=\epsilon\overline{\chi(-p)}$, Mellin inversion gives

$$\frac{1}{2\pi i}\int_{(c)}\Lambda_f(s,c_\chi)(\sqrt{M}y)^{1/2-s}\,ds=2y^{k/2}F_\chi(iy).$$

Use the functional equation and move the line from $1-c$ back to c to obtain

$$F_\chi(iy)=\lambda_\chi M^{-k/2}y^{-k}F_{\bar\chi}(i/(My)).$$

No residues occur because the completions are entire. Polynomial coefficient growth, Stirling's estimate on the two outer vertical lines, and the finite-order Phragmén–Lindelöf argument give exponential decay on the intervening closed strip, justifying the contour move. Replace y by $1/(My)$ and use analytic continuation in z to get (4). In particular, the $q=1$ case says

$$F|_k W_p=\eta F. \tag{5}$$

For a unit a modulo q, let $b(a)$ be the unit determined by $pab(a)\equiv -1 \pmod q$. Since

$$\chi(a)=\overline{\chi(-p)}\,\overline{\chi(b(a))},$$

equation (4) is

$$\sum_a \chi(a)\left(F|_k T_{a/q}W_{pq^2}-\eta F|_k T_{b(a)/q}\right)=0.$$

Orthogonality of all characters of $(\mathbb{Z}/q\mathbb{Z})^\times$ gives the individual equations

$$F|_k T_{a/q}W_{pq^2}=\eta F|_k T_{b/q}, \qquad pab\equiv -1 \pmod q. \tag{6}$$

Periodicity of F allows arbitrary integral lifts a,b. The formula remains valid for $q=1$.

Direct matrix multiplication gives

$$T_{a/q}W_{pq^2} = q\,\gamma_{q,a,b}W_pT_{b/q}. \tag{7}$$

Positive scalars act trivially, so (6), (7), and (5) imply

$$F|_k\gamma_{q,a,b} = F. \tag{8}$$

Proof of Theorem 1. The Fourier expansion gives $F|_kT_1 = F$. By Lemma 3, T_1, w_p, and the matrices in (8) generate G_p. The two cosets $\Gamma_0(p)$ and $\Gamma_0(p)w_p$ define a homomorphism from G_p to $\mathbb{Z}/2\mathbb{Z}$. Let ρ be the character which is 1 on the first coset and η on the second. This is a character because $\eta^2 = 1$. The transformation laws already proved say $F|_kg = \rho(g)F$ on every generator, and hence on all of G_p. In particular $F|_k\gamma = F$ for every $\gamma \in \Gamma_0(p)$. This also checks that a negative Fricke eigenvalue does not introduce a multiplier on $\Gamma_0(p)$.

For prime p, every cusp of $\Gamma_0(p)$ is equivalent to 0 or ∞. Indeed, for a reduced fraction a/c, if $p \mid c$ a Bezout completion with first column $(a, c)^t$ gives equivalence to ∞. If $p \nmid c$, solve $Ac - pBa = 1$ and use $\left(\begin{smallmatrix} A & a \\ pB & c \end{smallmatrix}\right)$ to obtain equivalence to 0. The positive Fourier expansion makes F vanish at ∞; (5) makes it vanish at 0. Hence it is a cusp form. □

3. The Euler-product version

Corollary 5. *Let $(f_n) \in \mathcal{E}(p, k)$, with untwisted root number ϵ. For every primitive Dirichlet character χ modulo $1 \le q < \sqrt{p}$, put*

$$\Lambda_f(s,\chi) = (pq^2)^{(s-1/2)/2}\Gamma_{\mathbb{C}}\left(s + \frac{k-1}{2}\right)\sum_{n\ge1}\frac{f_n\chi(n)}{n^s}, \qquad \tau(\chi) = \sum_{a \bmod q}\chi(a)e(a/q).$$

Suppose these functions are entire of finite order and satisfy

$$\Lambda_f(s,\chi) = \epsilon\chi(p)\frac{\tau(\chi)^2}{q}\Lambda_f(1-s,\bar\chi). \tag{9}$$

Then

$$F \in S_k^{\mathrm{new}}(\Gamma_0(p)) \subseteq M_k^\circ(\Gamma_0(p)).$$

Proof. For primitive χ, the identities $c_\chi(n) = \tau(\chi)\bar\chi(n)$ and $\tau(\chi)\tau(\bar\chi) = \chi(-1)q$ convert (9) into (1). By [1, Lemma 4.12], the unramified Hecke relations then give (1) for every character modulo every $q < \sqrt{p}$: the character-sum twist for an imprimitive character is its inducing primitive character-sum twist times a Dirichlet polynomial. The same identity preserves entireness and finite order. Thus Theorem 1 applies; polynomial coefficient growth follows from convergence of the original Dirichlet series in a half-plane. Write $a_n = f_nn^{(k-1)/2}$ and let the unnormalized operator U_p act by $\sum a_ne(nz) \mapsto \sum a_{pn}e(nz)$. The Euler factor gives $a_{pn} = \lambda a_n$ for every n, where

$$\lambda = p^{(k-1)/2}f_p, \qquad p^{1-k/2}\lambda = \sqrt{p}\,f_p \in \{1,-1\}.$$

The coset representatives I, ST_1^j $(0 \le j < p)$, with $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, give the usual trace identity; explicitly, $W_p\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right) = pST_1^j$ yields

$$\mathrm{Tr}^{\mathrm{SL}_2(\mathbb{Z})}_{\Gamma_0(p)}F = F + p^{1-k/2}U_p(F|_kW_p) = (1 + \eta\sqrt{p}\,f_p)F.$$

The scalar on the right is either 0 or 2. If it were 2, then F itself would be a level-one cusp form. In that case

$$F|_kW_p = F|_kS\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} = p^{k/2}F(pz),$$

which cannot equal $\eta F(z)$: the coefficient of $e(z)$ is zero on the left expression $p^{k/2}F(pz)$ and is $\eta a_1 = \eta \neq 0$ on the right. Thus the trace is zero. The trace of $F|_k W_p = \eta F$ is zero as well. These are precisely the two prime-level trace conditions defining the newspace in the main paper. □

Remark 6. The lower-bound examples of Steiner [2] explain the scale $\sqrt{p}$ for general linear converse theorems. They do not by themselves give a sharpness assertion within the Euler-product class $\mathcal{E}(p, k)$.

References

[1] S. Bettin, J. W. Bober, A. R. Booker, J. B. Conrey, M. Lee, G. Molteni, T. Oliver, D. J. Platt, and R. S. Steiner, *A conjectural extension of Hecke's converse theorem*, Ramanujan J. **47** (2018), 659–684, doi:10.1007/s11139-017-9953-y.

[2] R. S. Steiner, *Near counterexamples to Weil's converse theorem*, J. Théor. Nombres Bordeaux **31** (2019), 313–321, doi:10.5802/jtnb.1082.